

University of Amsterdam

MSc Computational Science Master Thesis

Thesis proposal

Nitai Nijholt

March 2, 2025

1. Title

Integrating LLM-as-a-judge and Context-Insensitive Hallucination reduction in LLM-Based Causal Loop Diagram Generation

2. Abstract

We propose a multi-agent LLM research system for generating initial Causal Loop Diagrams (CLDs), harnessing LLMs as low-cost information retrievers to address the knowledge synthesis challenges due to rapidly expanding research literature. Using generated initial, literature-grounded CLDs, domain experts can focus on adding specialized knowledge rather than reiterating consensus during model building sessions. To address LLM hallucinations threatening CLD accuracy, we integrate a context-sensitive verification approach (LLM-as-a-judge) with context-insensitive metrics (cosine similarity, perplexity), investigating whether applying these techniques—first separately, and then uniquely, in combination—can mitigate hallucinations, quantify uncertainty and increase final CLD accuracy more efficiently than regular test-time compute.

3. Research Context

Although LLM-based systems for CLD generation show promise to make expert model building sessions more efficient [7, 16, 3], LLM hallucinations—fabricated or inconsistent outputs—undermine reliability for scientific applications [20, 15]. Black-box mitigation strategies (prompting [18, 1], retrieval-augmented generation [5, 19], multi-agent orchestration [12, 8], and LLM-as-a-judge [21] and test-time compute [9, 12, 2]) and white-box approaches (fine-tuning [4], RLHF [10] activation monitoring [17]) have been found to reduce LLM hallucinations. LLM-as-a-judge advantage is context-awareness, but its vulnerable to (transformer-architectural) biases [6, 11]. We combine LLM-as-a-judge with a set of context-insensitive hallucination detection metrics (cosine similarity [5, 19], perplexity [14, 13], minimum probability [15])—an under-explored pairing that could allow for bias reduction and more efficient deployment of test-time compute for the system to self-correct hallucinations.

4. Research Methods

We design three complementary experiments to evaluate LLM-driven generation of Causal Loop Diagrams (CLDs). The experiments take place in the context of a CLD building system adhering implemented by Andrew Crossley, see in Table 1 for an overview of the algorithm. The system begins with a target variable and user-defined context, iteratively proposing new variables and pairwise relationships, each accompanied by an

LLM-generated rationale and cited source. We define a hallucination when the rationale contradicts itself, does not match its cited source, or fails user-specified criteria (e.g., causal relevance, measurability, non-overlapping). Final diagrams are verified against a separate, smarter reasoning LLM (Open AI o1 for example) or human experts in a later stage if feasible to reduce circularity (i.e. LLMs judging LLMs).

Research Question 1: “Can LLM-as-a-judge and LLM-as-corrector method be used to increase final CLD accuracy?”

- **RQ 1a:** “Can LLM-as-a-judge be used to detect hallucinations?”
- **RQ 1b:** “Can LLM-as-corrector be used to correct hallucinations?”

Experiment 1: LLM-as-a-judge.

To address RQ1, we select some known-to-be-correct ground truth CLDs, then create spurious versions by deliberately altering links and providing a spurious rationale. An LLM-as-a-judge checks each link’s rationale and source via which flags possible hallucinations. An “LLM-as-a-corrector” revises or removes flagged links. We compare final CLD accuracy and compute cost to a baseline with no verification. We expect the judge/corrector method to yield more accurate final CLDs.

Research Question 2: “Can context insensitive metrics be used to detect hallucinations?”

Experiment 2: Context-insensitive metrics.

We measure perplexity, minimum probability, and cosine similarity between the generated rationale and its cited text, correlating these metrics with flagged hallucinations. This indicates whether these low-compute cost, context-insensitive measures can effectively predict hallucinations and serve as a type of uncertainty quantification. We expect lower minimum probability and higher perplexity respectively to correspond to higher likelihood of hallucination and higher cosine similarity with (chunks) of the cited source to indicate lower likelihood of hallucination.

Research Question 3: “Can context-insensitive metrics when used as hallucination-indicators be used to deploy test-time compute to correct hallucinations more efficiently?”

Experiment 3: Combined approach.

We selectively apply LLM-as-a-judge and LLM-as-a-corrector only when context-insensitive metrics exceed certain thresholds—a “smart” test-time compute strategy. We compare final CLD accuracy and compute usage to an always-on approach (“dumb” test-time compute), which applies LLM-as-a-judge and LLM-as-a-corrector on every link and variable rationale, assessing whether targeted interventions reduce hallucinations more efficiently. We expect the smart deployment of test-time compute to increase efficiency and still meaningfully increase Final CLD accuracy.

5. Significance of Research

Uniquely combining LLM-as-a-judge with context-insensitive hallucination mitigation, our CLD building system generates self-corrected, uncertainty-quantified initial CLDs, redirecting experts attention in CLD-building session from reiterating consensus to adding specialized knowledge to models. Rigorous CLDs translate into coherent computational models for research and (intervention) simulation. Findings can reduce hallucinations and increase efficiency of test-time compute in multi-agent LLM-based research systems.

6. Bibliography

- [1] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.
- [2] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- [3] Niyousha Hosseinichimeh, Aritra Majumdar, Ross Williams, and Navid Ghaffarzadegan. From text to map: A system dynamics bot for constructing causal loop diagrams, 2024. URL <https://arxiv.org/abs/2402.11400>.
- [4] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- [5] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33:9459–9474, 2020.
- [6] Haitao Li, Qian Dong, Junjie Chen, Huixue Su, Yujia Zhou, Qingyao Ai, Ziyi Ye, and Yiqun Liu. Llms-as-judges: A comprehensive survey on llm-based evaluation methods. *arXiv preprint arXiv:2412.05579*, 2024.
- [7] Ning-Yuan Georgia Liu and David Keith. Leveraging large language models for automated causal loop diagram generation: Enhancing system dynamics modeling through curated prompting techniques. *SSRN Electronic Journal*, 2024. doi: 10.2139/ssrn.4906094. URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4906094.

- [8] C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune, and D. Ha. The ai scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint*, arXiv:2408.06292, 2024.
- [9] OpenAI. Learning to reason with llms: Introducing openai o1, a new large language model trained with reinforcement learning for complex reasoning. <https://openai.com/index/learning-to-reason-with-llms/>, 2024.
- [10] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- [11] Jiawen Shi, Zenghui Yuan, Yinuo Liu, Yue Huang, Pan Zhou, Lichao Sun, and Neil Zhenqiang Gong. Optimization-based prompt injection attack to llm-as-a-judge. In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*, pages 660–674, 2024.
- [12] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. *Advances in Neural Information Processing Systems*, 36, 2024.
- [13] SM Tonmoy, SM Zaman, Vinija Jain, Anku Rani, Vipula Rawte, Aman Chadha, and Amitava Das. A comprehensive survey of hallucination mitigation techniques in large language models. *arXiv preprint arXiv:2401.01313*, 2024.
- [14] Simon Valentin, Jinmiao Fu, Gianluca Detommaso, Shaoyuan Xu, Giovanni Zappella, and Bryan Wang. Cost-effective hallucination detection for llms. *arXiv preprint arXiv:2407.21424*, 2024. URL <https://arxiv.org/abs/2407.21424>.
- [15] Neeraj Varshney, Wenlin Yao, Hongming Zhang, Jianshu Chen, and Dong Yu. A stitch in time saves nine: Detecting and mitigating hallucinations of llms by validating low-confidence generation. *arXiv preprint arXiv:2307.03987*, 2023.
- [16] M. Waaijers, H. Rosenbusch, C. J. van Lissa, A. Roefs, and D. Borsboom. theoraizer: Ai-assisted theory construction. *PsyArXiv Preprints*, 2024. URL https://osf.io/preprints/psyarxiv/gu9yq_v1. *Preprint*.
- [17] T. Wang, X. Jiao, Y. He, Z. Chen, Y. Zhu, X. Chu, J. Gao, Y. Wang, and L. Ma. Adaptive activation steering: A tuning-free llm truthfulness improvement method for diverse hallucinations categories. *arXiv preprint*, 2024. URL <https://arxiv.org/html/2406.00034v1>.
- [18] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in

large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.

- [19] Yuzhe Zhang, Yipeng Zhang, Yidong Gan, Lina Yao, and Chen Wang. Causal graph discovery with retrieval-augmented generation based large language models. *arXiv preprint arXiv:2402.15301*, 2024. URL <https://arxiv.org/abs/2402.15301>.
- [20] Haochen Zhao, Hang Chen, Fan Yang, Ninghao Liu, Hang Deng, Hao Cai, Shuai Wang, Dong Yin, and Min Du. Explainability for large language models: A survey. *arXiv preprint arXiv:2309.01029*, 2023. URL <https://arxiv.org/abs/2309.01029>.
- [21] Lianmin Zheng, Wei-Lin Chiang, Yu Sheng, Shaohan Zhuang, Zhentao Wu, Yongkai Zhuang, Zhepei Lin, Zhuohan Li, Da Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *arXiv preprint arXiv:2306.05685*, 2023. URL <https://arxiv.org/abs/2306.05685>.

7. Tentative Timeline

Total Duration: 8 months effectively (9 months + 1 month break, leaving 1 month of grace period for unexpected delays)

January Started working on the Thesis Jan 21st. Started familiarizing with the codebase of the CLD system built by Andrew Crossley. First goal is to Submit thesis proposal draft to supervisor before end of January for feedback. In parallel conduct literature review.

February: Formalize and submit thesis proposal. Continue the literature review. Develop a minimum viable product (MVP) for the LLM-as-a-judge verification agent. Conduct initial experiments and interpret preliminary results using LLM-as-a-judge.

March: Expand LLM-as-a-judge experiments, develop different types of multi-agent orchestration/test-time compute setup (i.e. serial or parallel LLM calls).

April: Implement a set of context-insensitive metrics (MVP). Conduct initial experiments and interpret preliminary results. Based on findings, focus on refining the current method or exploring extensions proposed in the research methods section.

May: Develop experiments and MVP’s further, come up with suitable visualizations and include higher sample experiments. Start writing preliminary results.

June: Review additional literature and relate findings to preliminary results. Review more relevant literature and begin writing the discussion. (Contingent on progress) Collect expert validated dataset of reasoning traces leading to valid CLDs and finetune LLM on this data. Find out if CLD building accuracy improves as a result.

July: Continue writing discussion and refining writing of the thesis. (Contingent on progress) Explore white-box extensions (e.g., neuron activation analysis) or continue with

previous goals.

August: Summer break

September: Finalize writing. Submit thesis. Prepare for the defense.

Defense of thesis.

1 Appendix

1.1 Overview of Algorithm of the CLD building system

Step	Description
Initialization	• User adds single target variable around which the CLD is to be built, and 1) temporal 2) spatial and 3) other context to the prompt of the generating LLM. User also specifies a maximum number of variables. • After the user presses start, the target variable is as starting node to the CLD. Then the Variable generation loop starts:
Variable Generation Loop	1. Prompt the LLM for a new variable influencing the target. 2. Parse the response to extract the variable name/definition. 3. Insert a new variable node into the CLD. 4. Update the prompt so the LLM avoids duplicates. 5. Repeat until the desired number of variables is reached. 6. User then presses a button to assign causal relationships which triggers the Assign Relationship Loop.
Assign Relationships Loop (N-by-N)	1. Collect all CLD variables (N total). 2. For each pair (V_i, V_j), prompt the LLM to check if a <i>direct</i> causal link exists. 3. If it exists (REINFORCING, BALANCING, or INVERSE), add an edge with the specified polarity. 4. This yields up to $N \times N$ relationships to be tested. 5. Relationships and variables each have annotation containing LLMs rationale for adding it and scientific citations its based on.
Halting Condition	• All variable pairs are iterated over and relationships assigned, and the user specified specified number of variables has been hit.

Table 1: Algorithmic steps for iteratively building a Causal Loop diagram using the LLM-based CLD building system